VIT-AP UNIVERSITY

DIGITAL LOGIC DESIGN

(Course Code: ECE 1003)

Module-1:Lecture-4

Number Systems

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Module-1 (Part-1)

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- (b-1)'s Complement
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- b's Complement
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BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Number Systems

COMPLEMENT OF NUMBERS

- ✓ In general, **complements** are used to **simplify the subtraction operation and for logical manipulation**.
- ✓ There two types of complement method in any base-b number system.
  - **b's Complement (Radix Complement)**
  - **(b-1)'s Complement (Diminished Radix Complement)**
- ✓ In base-2 number system
  - 2's complement
  - 1's complement
- ✓ In base-10 number system
  - 10's complement
  - 9's complement

DIMINISHED RADIX [(b-1)'s] COMPLEMENT

- ✓ If a number ***N*** having ***n*** digits with base ***b***, then the (b-1)'s complement is given by

$$(b^n - 1) - N$$

- ✓ For **binary number**, it is **1's complement** and can be represented as

$$(2^n - 1) - N$$

- ✓ For Example: 1's complement of $\mathbf{(10101)_2}$

$$(2^5 - 1) - (10101)_2$$

$$\Rightarrow (32 - 1) - (10101)_2$$

$$\Rightarrow (31) - (10101)_2 \qquad \Rightarrow (11111)_2 - (10101)_2 = (01010)_2$$

DIMINISHED RADIX [(b-1)'s] COMPLEMENT

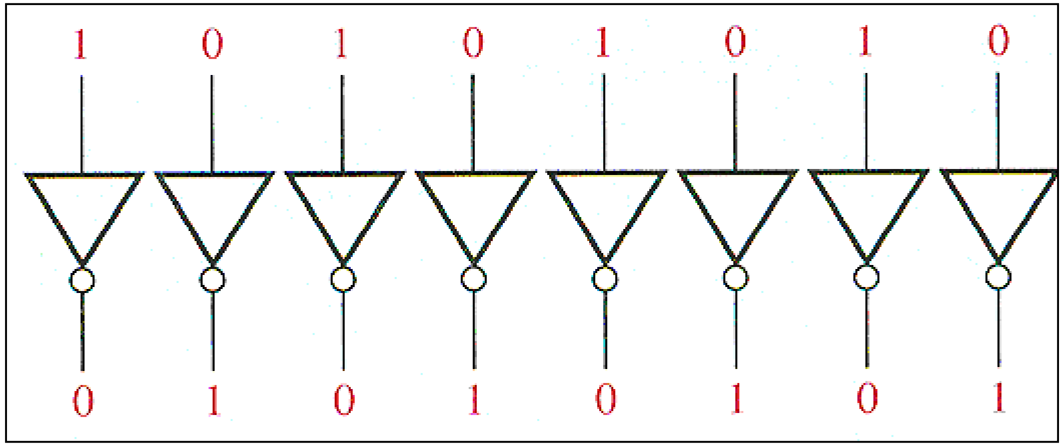
Note:

- ✓ **The 1's complement of a binary number can be obtained by changing each 0's to 1's and each 1's to 0's.**
- ✓ For example: 1's complement of $\mathbf{(11101)_2}$

1 1 1 0 1

↓ ↓ ↓ ↓ ↓

0 0 0 1 0



1's complement of $\mathbf{(11101)_2} = \mathbf{(00010)_2}$

DIMINISHED RADIX [(b-1)'s] COMPLEMENT

- ✓ For **decimal number**, it is **9's complement** and can be represented as

$$(10^n - 1) - N$$

- ✓ For Example: 9's complement of $\mathbf{(54230)_{10}}$

$$(10^5 - 1) - (54230)_{10}$$

$$\Rightarrow (100000 - 1) - (54230)$$

$$\Rightarrow (99999) - (54230)$$

$$\Rightarrow 45769$$

DIMINISHED RADIX [(b-1)'s] COMPLEMENT

Note:

- ✓ **The 9's complement of a decimal number can be obtained by subtracting each digit by 9.**
- ✓ For example: 9's complement of $(\mathbf{12345})_{10}$

$$\begin{array}{r} 9\;9\;9\;9\;9 \\ -\;1\;2\;3\;4\;5 \\ \hline 8\;7\;6\;5\;4 \end{array}$$

9's complement of $(\mathbf{12345})_{10} = (\mathbf{87654})_{10}$

RADIX (b's) COMPLEMENT

- ✓ If a number ***N*** having ***n*** digits with base ***b***, then the b's complement is given by

$$b^n - N = [(b^n - 1) - N] + 1 = \text{(b-1)'s Complement + 1}$$

- ✓ For **binary number**, it is **2's complement** and can be represented as

$$(2^n - N)$$

- ✓ For Example: 2's complement of $\mathbf{(10111)_2}$

$$2^5 - (10111)_2$$

$$\Rightarrow 32 - (10111)_2$$

$$\Rightarrow (100000)_2 - (10111)_2 \quad \Rightarrow (01001)_2$$

1's complement of $\mathbf{(10111)_2}$ = $\mathbf{(01000)_2}$

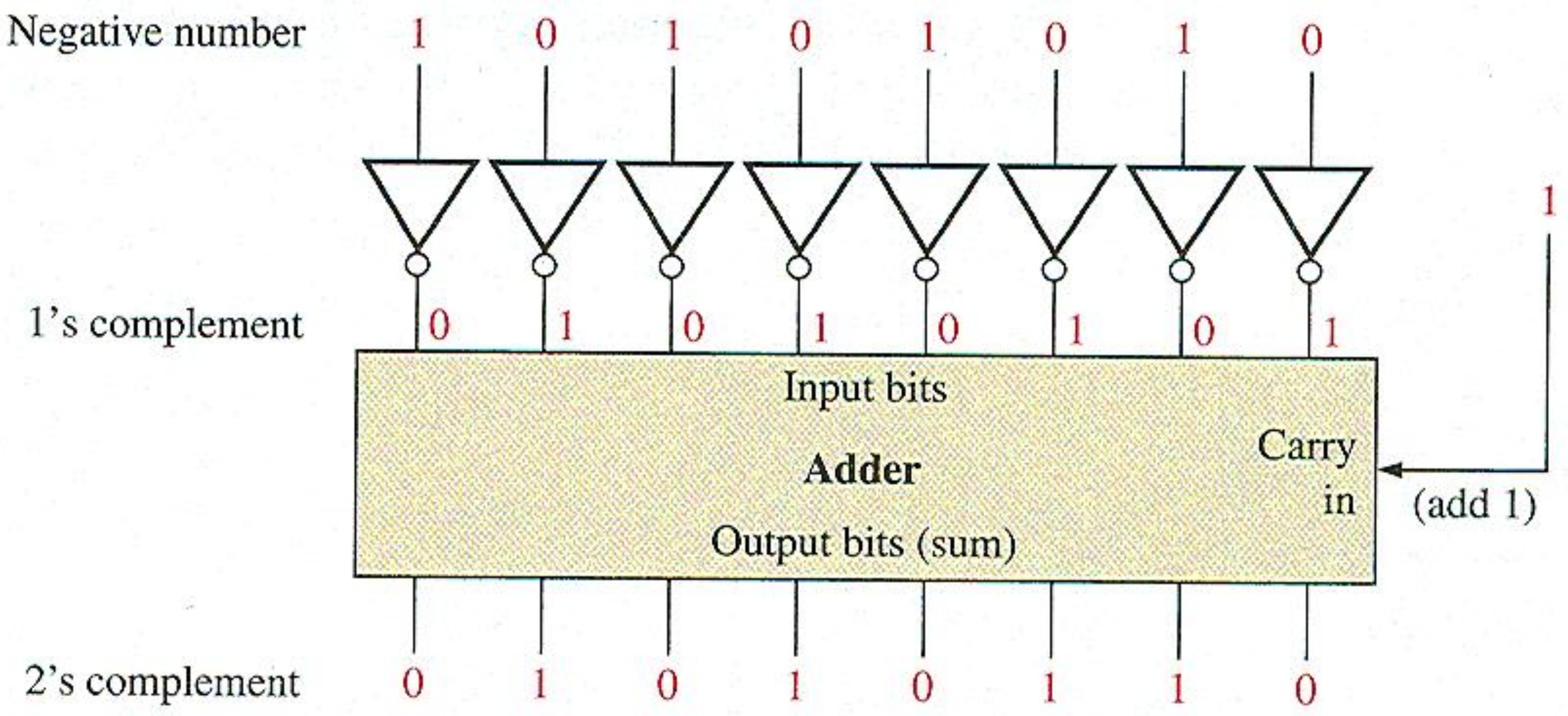
2's complement of $\mathbf{(10111)_2}$

= 1's complement of $\mathbf{(10111)_2}$ **+ 1**

= $\mathbf{(01000)_2}$ **+ 1** = $\mathbf{(01001)_2}$

RADIX (b's) COMPLEMENT

✓ For **binary number**, it is **2's complement** and can be represented as



RADIX (b's) COMPLEMENT

Note:

✓ **The 2's complement of a binary number can be obtained by leaving the least significant 0's and the first 1 unchanged and then remaining bits, replace 1's with 0's and 0's with 1's.**

✓ For example: 2's complement of $(\mathbf{1110100})_2$

$$1\ 1\ 1\ 0\ \underbrace{1\ 0\ 0}_{\text{unchanged}}$$

$$\downarrow\ \downarrow\ \downarrow\ \downarrow\ \downarrow\ \downarrow\ \downarrow$$

$$0\ 0\ 0\ 1\ 1\ 0\ 0$$

2's complement of $(\mathbf{1110100})_2 = (\mathbf{0001100})_2$

2's complement of $(\mathbf{100000})_2$

$$\underbrace{1\ 0\ 0\ 0\ 0\ 0}_{\text{unchanged}}$$

$$\downarrow\ \downarrow\ \downarrow\ \downarrow\ \downarrow\ \downarrow$$

$$1\ 0\ 0\ 0\ 0\ 0$$

2's complement of $(\mathbf{100000})_2 = (\mathbf{100000})_2$

RADIX (b's) COMPLEMENT

- For **decimal number**, it is **10's complement** and can be represented as

$$(10^n - N)$$

- For Example: 10's complement of $\mathbf{(54230)_{10}}$

$$10^5 - (54230)_{10}$$

$$\Rightarrow (100000) - (54230)$$

$$\Rightarrow 45770$$

9's complement of $\mathbf{(54230)_{10}}$ = $\mathbf{(45769)_{10}}$

10's complement of $\mathbf{(54230)_{10}}$

= 9's complement of $\mathbf{(54230)_{10} + 1}$

= $\mathbf{(45769)_{10} + 1 = (45770)_{10}}$

RADIX (b's) COMPLEMENT

Note:

- ✓ **The 10's complement of a decimal number can be obtained by leaving the least significant 0's unchanged and the first least non-zero element should be subtracted from 10 and other digits from 9.**
- ✓ For example: 10's complement of **$(2860010)_2$**

$$2\ 8\ 6\ 0\ 0\ 1\ \underbrace{0}_{\text{unchanged}}$$

$$\begin{array}{r} 9\ 9\ 9\ 9\ 9\ 10\ 0 \\ -\ 2\ 8\ 6\ 0\ 0\ 1 \\ \hline 7\ 1\ 3\ 9\ 9\ 9\ 0 \end{array}$$

10's complement of **$(2860010)_{10}$ = $(7139990)_2$**

SUBTRACTION WITH COMPLEMENT

- ✓ In digital hardware, complement method for subtraction provides more efficient results compared to borrow concept method.
- ✓ Here, the subtraction is pursued for unsigned numbers using the following method
  - **Subtraction of unsigned numbers with b's Complement**
  - **Subtraction of unsigned numbers (b-1)'s Complement**

SUBTRACTION WITH b's COMPLEMENT

Subtraction of unsigned numbers with b's Complement

- ✓ The subtraction of **two n-digit unsigned numbers M - N** in base **b** can be done as follows:

1. Add the minuend **M** to the b's complement of the subtrahend **N**. Mathematically,

$$M + (b^n - N) = M - N + b^n$$

2. If $M \geq N$, the sum will produce an end carry b^n, which can be discarded; what is left is the result $M - N$.

3. If $M < N$, the sum does not produce an end carry and is equal to $b^n - (N - M)$, which is the b's complement of $(N - M)$. To obtain the answer in a familiar form, take the b's complement of the sum and place a negative sign in front.

SUBTRACTION WITH b's COMPLEMENT

Example-1:

✓ Using 2's complement, subtract $(\mathbf{101111})_2 - (\mathbf{100110})_2$.

Solution: Here, **M = $(101111)_2$** and **N = $(100110)_2$**

2's complement of **N = $(011010)_2$**

Step -1: Add M with 2's complement of N

```
   1   1   1   1   1
       1   0   1   1   1   1
+      0   1   1   0   1   0
---------------------------
   1   0   0   1   0   0   1
```

Discard End Carry

Step -2: Here M>N, hence, the result is $(\mathbf{101111})_2 - (\mathbf{100110})_2 = (001001)_2$

SUBTRACTION WITH b's COMPLEMENT

Example-2:

✓ Using 2's complement, subtract $(100)_2 - (110000)_2$.

Solution: Here, $M = (100)_2$ and $N = (110000)_2$

2's complement of $N = (010000)_2$

Step -1: Add M with 2's complement of N

$$\begin{array}{rcccccc} & 0 & 0 & 0 & 1 & 0 & 0 \\ + & 0 & 1 & 0 & 0 & 0 & 0 \\ \hline & 0 & 1 & 0 & 1 & 0 & 0 \end{array}$$

Step -2: Here M<N, hence, the result is obtained by taking the 2's complement of sum with a negative sign in front.

$$(100)_2 - (110000)_2 = -[\text{2's complement of } (010100)_2] = -(101100)_2$$

SUBTRACTION WITH b's COMPLEMENT

Example-3:

✓ Using 10's complement, subtract **52532 – 3250**.

Solution: Here, **M = 52532** and **N = 3250**

10's complement of **N = (99999 – 03250) + 1**

= 96749 + 1 = 96750

Step -1: Add M with 10's complement of N

$$
\begin{array}{cccccc}
1 & & 1 & & & \\
 & 5 & 2 & 5 & 3 & 2 \\
+ & 9 & 6 & 7 & 5 & 0 \\
\hline
1 & 4 & 9 & 2 & 8 & 2
\end{array}
$$

Discard End Carry

Step -2: Here M>N, hence, the result is **52532 – 3250 = 49282**

SUBTRACTION WITH b's COMPLEMENT

Example-4:

✓ Using 10's complement, subtract **1020 – 2056**.

Solution: Here, **M = 1020** and **N = 2056**

10's complement of **N = (9999 – 2056) + 1**

= 7943 + 1 = 7944

```
    1  0  2  0
+   7  9  4  4
---------------
    8  9  6  4
```

Step -1: Add M with 10's complement of N

Step -2: Here M<N, hence, the result is obtained by taking the 10's complement of sum with a negative sign in front.

1020 – 2056 = –[10's complement of 8964] = –[(9999 – 8964) + 1] = – [1035 + 1] = – 1036

SUBTRACTION WITH (b-1)'s COMPLEMENT

Subtraction of unsigned numbers with (b-1)'s Complement

✓ The subtraction of **two n-digit unsigned numbers M - N** in base **b** can be done as follows:

1. Add the minuend **M** to the (b-1)'s complement of the subtrahend **N**. Mathematically,

$$M + ((b^n - 1) - N) = M - N + b^n - 1$$

2. If $M \geq N$, the sum will produce an end carry b^n. To obtain the result, remove the end carry and add 1 to the sum which also termed as **end around carry**.

3. If $M < N$, the sum does not produce any carry. To obtain the answer in a familiar form, take the (b-1)'s complement of the sum and place a negative sign in front.

SUBTRACTION WITH (b-1)'s COMPLEMENT

Example-1 :

✓ Using 1's complement, subtract $(101111)_2 - (100110)_2$.

Solution: Here, $M = (101111)_2$ and $N = (100110)_2$

1's complement of $N = (011001)_2$

Step -1: Add M with 1's complement of N

```
   1   1   1   1   1   1
       1   0   1   1   1   1
+      0   1   1   0   0   1
----------------------------
   1   0   0   1   0   0   0
+                          1
----------------------------
       0   0   1   0   0   1
```

End Around Carry

Step -2: Here M>N, hence, the result is $(101111)_2 - (100110)_2 = (001001)_2$

SUBTRACTION WITH (b-1)'s COMPLEMENT

Example-2:

✓ Using 1's complement, subtract $(100)_2 - (110000)_2$.

Solution: Here, $M = (100)_2$ and $N = (110000)_2$

1's complement of $N = (001111)_2$

Step -1: Add M with 1's complement of N

$$
\begin{array}{ccccccc}
 & & 1 & 1 & & & \\
 & 0 & 0 & 0 & 1 & 0 & 0 \\
+ & 0 & 0 & 1 & 1 & 1 & 1 \\
\hline
 & 0 & 1 & 0 & 0 & 1 & 1
\end{array}
$$

Step -2: Here M<N, hence, the result is obtained by taking the 1's complement of sum with a negative sign in front.

$$(100)_2 - (110000)_2 = -[\text{1's complement of } (010011)_2] = -(101100)_2$$

SUBTRACTION WITH (b-1)'s COMPLEMENT

Example-3:

✓ Using 9's complement, subtract **52532 – 3250**.

Solution: Here, **M = 52532** and **N = 3250**

9's complement of **N = (99999 – 03250)**

= 96749

Step -1: Add M with 9's complement of N

$$
\begin{array}{cccccccc}
 & 1 & & 1 & & 1 & & \\
 & & 5 & 2 & 5 & 3 & 2 \\
+ & & 9 & 6 & 7 & 4 & 9 \\
\hline
 & 1 & 4 & 9 & 2 & 8 & 1 \\
+ & & & & & & 1 \\
\hline
 & & 4 & 9 & 2 & 8 & 2
\end{array}
$$

End Around Carry

Step -2: Here M>N, hence, the result is **52532 – 3250 = 49282**

SUBTRACTION WITH (b-1)'s COMPLEMENT

Example-4:

✓ Using 9's complement, subtract **1020 – 2056**.

Solution: Here, **M = 1020** and **N = 2056**

9's complement of **N = (9999 – 2056)**

= 7943

$$\begin{array}{rrrrr} & 1 & 0 & 2 & 0 \\ + & 7 & 9 & 4 & 3 \\ \hline & 8 & 9 & 6 & 3 \end{array}$$

Step -1: Add M with 9's complement of N

Step -2: Here M<N, hence, the result is obtained by taking the 9's complement of sum with a negative sign in front.

1020 – 2056 = –[9's complement of 8963] = –(9999 – 8963) = – 1036

ASSIGNMENT QUESTIONS

1. Using 1's, 2's, 9's and 10's complement perform the following

   (a) 23-56

   (b) 98-34


Thank You